

SYNTHESIS

# Ecological niche conservatism: a time-structured review of evidence

A. Townsend Peterson

Biodiversity Institute, University of Kansas,  
Lawrence, KS 66045, USA

## ABSTRACT

**Aim** To evaluate the evolutionary conservatism of coarse-resolution Grinnellian (or scenopoetic) ecological niches.

**Location** Global.

**Methods** I review a broad swathe of literature relevant to the topic of niche conservatism or differentiation, and illustrate some of the resulting insights with exemplar analyses.

**Results** Ecological niche characteristics are highly conserved over short-to-moderate time spans (i.e. from individual life spans up to tens or hundreds of thousands of years); little or no ecological niche differentiation is discernible as part of the processes of invasion or speciation.

**Main conclusions** Although niche conservatism is widespread, many methodological complications obscure this point. In particular, niche models are frequently over-interpreted: too often, they are based on limited occurrence data in high-dimensional environmental spaces, and cannot be interpreted robustly to indicate niche differentiation.

## Keywords

Ecological niche, evolutionary change, Grinnell, niche conservatism, niche identity, niche similarity, overfitting, speciation, temporal scale.

Correspondence: A. Townsend Peterson,  
Biodiversity Institute, University of Kansas,  
Lawrence, KS 66045, USA.  
E-mail: town@ku.edu

## INTRODUCTION

The idea of ecological niche conservatism was proposed explicitly and tested quantitatively only 10 years ago (Peterson *et al.*, 1999), although earlier workers had approached the issue (Huntley *et al.*, 1989; Ricklefs & Latham, 1992; Peterson & Vargas-Barajas, 1993), with the conclusion that ecological niches show considerable conservatism over evolutionary time periods. Subsequent discussions and reviews of the subject (Ackerly, 2003; Wiens & Graham, 2005; Losos, 2008; Pearman *et al.*, 2008; Warren *et al.*, 2008; Wiens, 2008) have shed new light on the issue, particularly as additional evidence has accumulated. For example, Wiens & Graham (2005) pointed out the many dimensions in which niche conservatism has pervasive influences across evolutionary biology and biogeography. Pearman *et al.* (2008) emphasized topics of overlap between ecological and phylogenetic studies. Warren *et al.* (2008), in the most fundamentally important of these papers, pointed out that two distinct null hypotheses had been tested in previous studies (i.e. testing the hypothesis that two modelled niches are identical versus the hypothesis that two

modelled niches are more similar than would be expected at random), which probably explains a substantial part of the divergence among results.

However, these discussions have obfuscated key aspects of the niche conservatism issue as well. While I agree with Wiens & Graham (2005) that simply testing for conservatism in and of itself is not of particular interest, I see the challenge of understanding and characterizing the temporal dimensions of ecological innovation as being of great interest. This temporal point of view has been lacking in reviews of niche conservatism to date, so I believe that a reconsideration of the question of niche conservatism across different time-scales is in order, and this indeed is the purpose of the present paper.

## REVIEW

Since the initial publications regarding niche conservatism, the idea has attracted quite a bit of attention, and the body of relevant literature has grown considerably. As such, after a decade, it is useful to step back and ask where the field stands. The key aspect of this review is that I have organized available

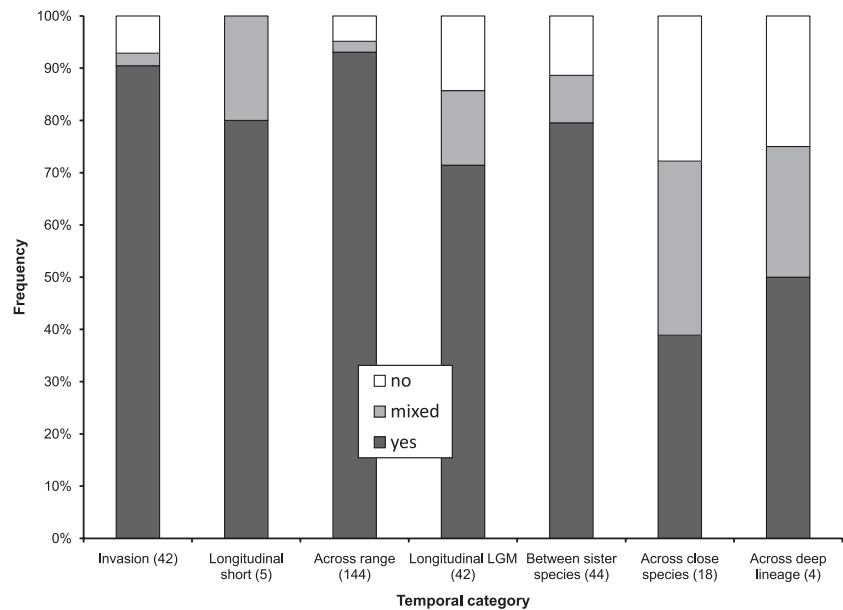
**Table 1** Summary of temporal dimensions of evidence regarding ecological niche conservatism. Relevant phenomena are classified according to what factor changes for the lineage in question, the approximate time-scale of these changes, and key details (with example references) regarding such evidence.

| Phenomenon                                        | What changes                                            | Approximate time-scale | Notes regarding review of evidence                                                                                                                                                                                     |
|---------------------------------------------------|---------------------------------------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Species invasions                                 | Landscape, community context                            | $10^0$ – $10^2$ years  | Requires independent testing, with models built on one distributional area and projected to and tested for validity on another (Peterson, 2003)                                                                        |
| Longitudinal: short-term shifts of species ranges | Conditions                                              | $10^1$ – $10^2$ years  | Requires independent samples of a species' distributional area before and after a documented environmental change event (Araújo <i>et al.</i> , 2005)                                                                  |
| Geographical distributions                        | Landscape                                               | $10^2$ – $10^5$ years  | Requires spatially stratified testing of model validity to establish consistency of ecological niche characteristics across a species' range (Peterson & Holt, 2003)                                                   |
| Longitudinal: shifts from Pleistocene to present  | Conditions                                              | $10^5$ – $10^6$ years  | Requires independent samples of a species' distributional area in the Pleistocene and at present (Martínez-Meyer & Peterson, 2006), or genetic evidence as a surrogate for palaeodistribution (Peterson & Nyári, 2007) |
| Comparisons between sister species                | Speciation event, often different landscape, conditions | $10^5$ – $10^6$ years  | Requires simple phylogenetic hypotheses indicating likely sister lineages (Peterson <i>et al.</i> , 1999)                                                                                                              |
| Across phylogeny – close species                  | Landscape, speciation events, conditions                | $10^5$ – $10^7$ years  | Requires phylogenetic hypothesis for monophyletic lineage of closely related species (Rice <i>et al.</i> , 2003; Knouft <i>et al.</i> , 2006)                                                                          |
| Across phylogeny – deeper                         | Landscape, speciation events, conditions                | $10^6$ – $10^7$ years  | Requires phylogenetic hypothesis for monophyletic lineages of species (Eaton <i>et al.</i> , 2008)                                                                                                                     |

evidence by temporal scale (Table 1). I cover a broader range of literature than previous reviews, including evidence from studies on topics as diverse as species invasions, geographically structured distributional predictions, longitudinal studies (i.e. studies examining a single species or lineage through time) and phylogenetic studies that consider explicitly the evolutionary history of ecological characteristics. In choosing studies for inclusion in this overview, I searched Google Scholar (<http://scholar.google.com/>) using the keywords 'niche' and 'distribution model', as well as (separately from keyword searches) the names of leading researchers in this field (e.g. R. Anderson, M. Araújo, C. Graham, A. Guisan, J. Lobo, E. Martínez-Meyer, R. Pearson, J. Soberón, W. Thuiller). I included only studies that met the following basic requirements: (1) ecological requirements were evaluated across significant portions of ranges of species (i.e. excluding studies that evaluated niches based on small and potentially non-representative subsets of species distributional areas, such as Vanreusel *et al.*, 2007); (2) spatial or temporal stratification of training and testing data sets was employed, thus avoiding problems with autocorrelation and non-independence of training and testing samples (eliminating, for example, Elith *et al.*, 2006); and (3) the focus was on coarse-resolution variables (i.e. the 'scenopoetic' variables of Hutchinson; see Soberón, 2007), rather than on more micro-scale environmental features that tend to incorporate elements of biotic interactions (eliminating, for example, Losos *et al.*, 2003). More generally, any such broad review,

and particularly one in which re-analysis of actual original data is not feasible, will necessarily smooth over many methodological differences and inconsistencies among studies – for example, in some summary statistics, I consider the relatively few studies testing niche identity together with studies testing niche similarity (Warren *et al.*, 2008), but checked the effects of their inclusion. I undertook the literature search in December 2008. While I have made every effort to make the literature review comprehensive, without doubt several relevant publications have escaped me, particularly given the burgeoning literature in the field. As such, this review serves as an overview of broad patterns only.

My review of the evidence for or against ecological niche conservatism covers a broader swathe of literature (76 publications) than previous reviews, and covers 299 species or higher taxa, again a much greater diversity of examples than previously analysed (see Appendix S1 in Supporting Information). On the most basic level, the results confirm the conclusions of all previous reviews: that evidence for niche conservatism is mixed. However, when sorting the evidence by time-scale, some degree of structure is evident in the results: recent and short-term events (e.g. species invasions, distributional shifts at the end of the Pleistocene) show a considerable tendency towards conservatism. Longer-term events, on the other hand, such as differentiation across phylogenies, show increasing degrees of breakdown of conservatism (Fig. 1). This trend in the prevalence of niche conservatism makes the mixed



**Figure 1** Summary of the review of evidence for ecological niche conservatism for various time-scales (see Table 1). Percentages reported on the vertical axis reflect numbers of species or higher taxa or of published studies, when individual lineage results were not reported (see Appendix S1).

and possibly confusing literature on the subject more intelligible: quite simply, and to many not surprisingly, niche conservatism breaks down over time.

A more important question is the rate at which this breakdown occurs. It is worthwhile to return to the issue on which the original niche conservatism analysis focused (Peterson *et al.*, 1999) – that of possible associations between speciation events and ecological innovation. What we can see in the results of this review, however, is that niche conservatism extends deeper in time than has generally been appreciated. That is, on time-scales more or less comparable to those of speciation events, almost all lineages show overall niche conservatism (Fig. 1). No ecological ‘signal’ associated with speciation is discernible. Indeed, if ecological innovation is at all involved in the speciation process, it must be manifested in fine-scale dimensions only: coarse-grained, range-wide-extent studies such as those reviewed herein do not show a signal of ecological differentiation associated with speciation events. Note that these patterns of overall conservatism on short-to-medium time-scales actually became stronger when six studies that used a distinct means of testing for conservatism (niche-identity tests, discussed in the next section) were removed from analysis, as all six found non-conservatism, and all were at the deeper-time end of the spectrum (see Appendix S1). As a consequence, my results should be robust to controlling for the diverse methods considered for testing conservatism.

## METHODOLOGICAL CONSIDERATIONS

The review just presented indicated a dominant pattern of minimal change in ecological niche parameters over short-to-medium time spans. In this section, I examine a few apparent exceptions in greater detail. First, however, a few more general methodological comments are in order.

Methodologies employed varied quite widely among the studies that met the criteria for inclusion in the review. The

simplest studies were spatially stratified prediction exercises, in which models trained in one part of a species’ geographical distribution were used to predict the species’ geographical distribution elsewhere (e.g. Peterson & Shaw, 2003). Studies covering a broader temporal swathe, however, either predicted among two temporal samples (‘before’ and ‘after’ some change event), or assessed coincidence and consistency of ecological characteristics within historical (phylogenetic) frameworks. Hence, certainly, some noise is inserted into the picture by these methodological variations among studies.

Because the field is so young, no synthesis of techniques and approaches has yet been published, and methodologies are evolving even from month to month: basically, investigators are exploring and experimenting in each publication that appears. As such, direct comparisons among studies and across taxa, regions or biomes are complicated for methodological and artefactual reasons, as has been pointed out recently in several publications (Peterson *et al.*, 2007, 2008; Lobo *et al.*, 2008; Warren *et al.*, 2008). Additional methodological considerations exist that can confuse the interpretation of such studies, as described below.

## Controlling overfitting and other artefacts

Three recent studies of native and introduced distributional areas of invasive species (Broennimann *et al.*, 2007; Fitzpatrick *et al.*, 2007; Medley, 2010) have reported niche shifts associated with the invasion event: in the European plant *Centaurea maculosa*, the South American fire ant *Solenopsis invicta*, and the mosquito *Aedes albopictus*, respectively. Although all three studies present intriguing evidence, interpreted by their authors as suggesting a rapid change of niche characteristics associated with invasion events, their inferences of niche shifts are dependent on the particular environmental data sets employed. In the case of the ant study, Peterson & Nakazawa (2008) developed additional niche models for the same

situations, and demonstrated that native range models for this species predicted the North American introduced range quite accurately when simpler and less highly dimensional environmental data sets were used (*contra* Fitzpatrick *et al.*, 2007): in sum, the ant example of niche shifts appears to be a consequence of methodological artefact, rather than a biological reality. A similar case study of the complex effects of using different sets of environmental variables on perceived niche 'differentiation' in Mediterranean house geckos (*Hemidactylus turcicus*) is provided by Rödder & Lötters (2009).

In the case of the *Centaurea* study by Broennimann *et al.* (2007), two subsequent publications (Broennimann & Guisan, 2008; Pearman *et al.*, 2008) have repeated the original assertions. However, in the supplementary material of Broennimann *et al.* (2007) it can be seen that under simpler sets of environmental data, the coincidence between the reciprocal predictions (native-to-introduced and vice versa) improves markedly. Put simply, the pattern parallels closely that of the *Solenopsis* situation detailed above: the results that are indicative of niche shifts are not robust to choice of environmental data, and the separation in the multivariate visualizations of niche space decreases when the model is built in fewer environmental dimensions. The mosquito example (Medley, 2010) is almost exactly parallel – niche models developed in a highly dimensional environmental space led to conclusions of niche shift, in this case in spite of a detailed previous analysis that showed clear niche conservatism (Benedict *et al.*, 2007). I do not argue necessarily that the conclusions of niche shifts are incorrect, but that the evidence for those conclusions should be more robust to alternative means of analysis before biological interpretation.

In all three examples, the authors complemented the niche model results by means of simple plots of the environmental values associated with known occurrences of each species, showing again – at least according to the authors' interpretations – that they are not equivalent. This evidence can appear quite convincing. However, these tests of niche identity are themselves at the centre of a knot of serious problems; in this respect, the reader should note the discussion of niche-identity tests presented below.

A more recent suggestion by two of the authors of the original *Centaurea* paper has been that of pooling native and introduced distributional data to produce a consensus model (Broennimann & Guisan, 2008). While in general I agree with this idea, that is, of bringing all of the available distributional information to bear on the challenge of predicting distributional potential, I note that *if* the assertion of niche differentiation were to be correct, this approach would then not be appropriate. Rather, the pooled niche model would be overly broad, and would not be representative of the ecological niche of either of the component populations (Raxworthy *et al.*, 2007).

### Managing dimensionality

One of the principal problems in the question of niche conservatism is that the answer to the question is dimension-

dependent. That is, if 'niche' is defined *very* simply, then niches will be conserved quite commonly. Consider, for example, characterizing species along a single environmental dimension such as annual mean temperature – niches would (in this one dimension) appear to be conserved massively, with examples of niche shifts being relatively rare. Clearly, this single dimension is not sufficient to summarize ecological variation among all of the species on Earth, so investigators must seek a richer characterization of environmental variation across regions of interest. This 'correct' number of dimensions is unknown, and we can discover it only indirectly.

On the other hand, niche models developed across too many dimensions fall into the opposite trap. If models are developed in 100 or 1000 rich environmental dimensions, then a characterization (model) can probably be developed that identifies each occurrence point uniquely. Such models will be massively overfitted (Peterson, 2007) and will have little or no predictive power, particularly in situations requiring that models be transferable (Peterson *et al.*, 2007). The conclusion in such analyses, by investigators not taking steps to avoid overfitting, will be niche differentiation.

Many such analyses have been based on 19 climatic dimensions (e.g. Broennimann *et al.*, 2007), but what is special or sacred about 19? In truth, the number comes from the fact that the WorldClim archive conveniently provides 19 'bioclimatic' variables (Hijmans *et al.*, 2005). At the very least, the dimensionality of the model needs to be stated and pondered carefully at the outset of the analysis; better still, investigators should test the appropriate dimensionality, given the species in question, the sample size of occurrence data available, and the particular set of environmental variables in question and their intercorrelations. The reduction of dimensionality for niche modelling as a step towards avoiding overfitting can be achieved: (1) via factor analysis to produce a reduced suite of independent dimensions for modelling (Peterson, 2007); or (2) via inspection of correlations among dimensions and the elimination of highly correlated dimensions from further analysis (Peterson, 2007). In the first case some degree of interpretability is lost, whereas in the second case some degree of subjectivity is involved.

### Non-identity versus niche differentiation

Warren *et al.* (2008) clarified that investigators in the field of niche modelling have been testing disparate null hypotheses, all under the rubric of 'niche differentiation', and provided randomization tests appropriate for each. In summarizing their re-analysis of two previous studies, they stated that 'our results strongly support the conventional wisdom that niches of related species tend to be similar, but rarely identical ... indeed ... our more stringent comparisons ... show that environmental niches can be quite labile' (p. 2877). However, Warren and colleagues perhaps did not consider in sufficient depth the effects of vagaries of sampling, model fitting and limited environmental availability in their tests of niche identity. This question is basically one of statistical power in

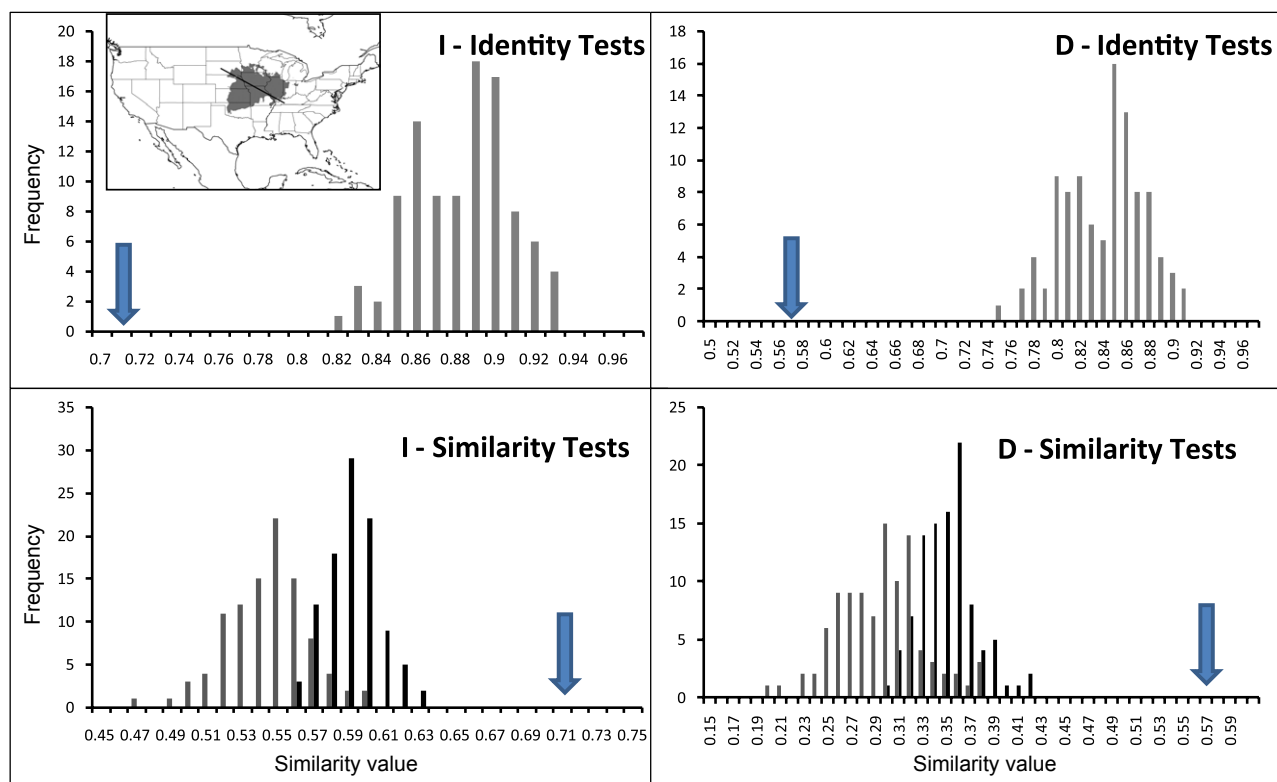
their tests: tests of niche identity may be overly prone to committing Type 1 errors, that is, of rejecting the null hypothesis of niche identity when such is nonetheless the case. The question thus arises of the degree to which statistically significant niche non-identity can or should be taken as an indication of *biologically significant* niche differentiation.

To explore these questions, at least heuristically, I created a virtual species following Hirzel *et al.* (2001) in a 7-dimensional climate space using bioclimatic variables from the WorldClim archive (Hijmans *et al.*, 2005), in which each dimension had been standardized to a mean of zero and variance of unity. I took as the centre of the virtual species' niche the conditions represented at Lawrence, Kansas, measuring the Euclidean distance in the standardized, 7-dimensional space (number of dimensions chosen arbitrarily) to all other points across North America. I arbitrarily chose a threshold of distance that included 5% of the North American environmental space to delimit this niche, and identified all sites across the continent that presented distances to Lawrence's conditions less than or equal to the threshold distance as the potential geographical distribution of the niche. Finally, using a biogeographer's eye, I split this potential distribution into two virtual species' distributions based approximately on the Missouri River as a biogeographical barrier (Fig. 2), and then plotted 20 points at random across each of the two distributional areas. The point

of this exercise is that points analogous to occurrence points for real species were chosen from distributional areas that were outlined by a single ecological niche defined in seven scenopoetic climatic dimensions (Soberón & Peterson, 2005).

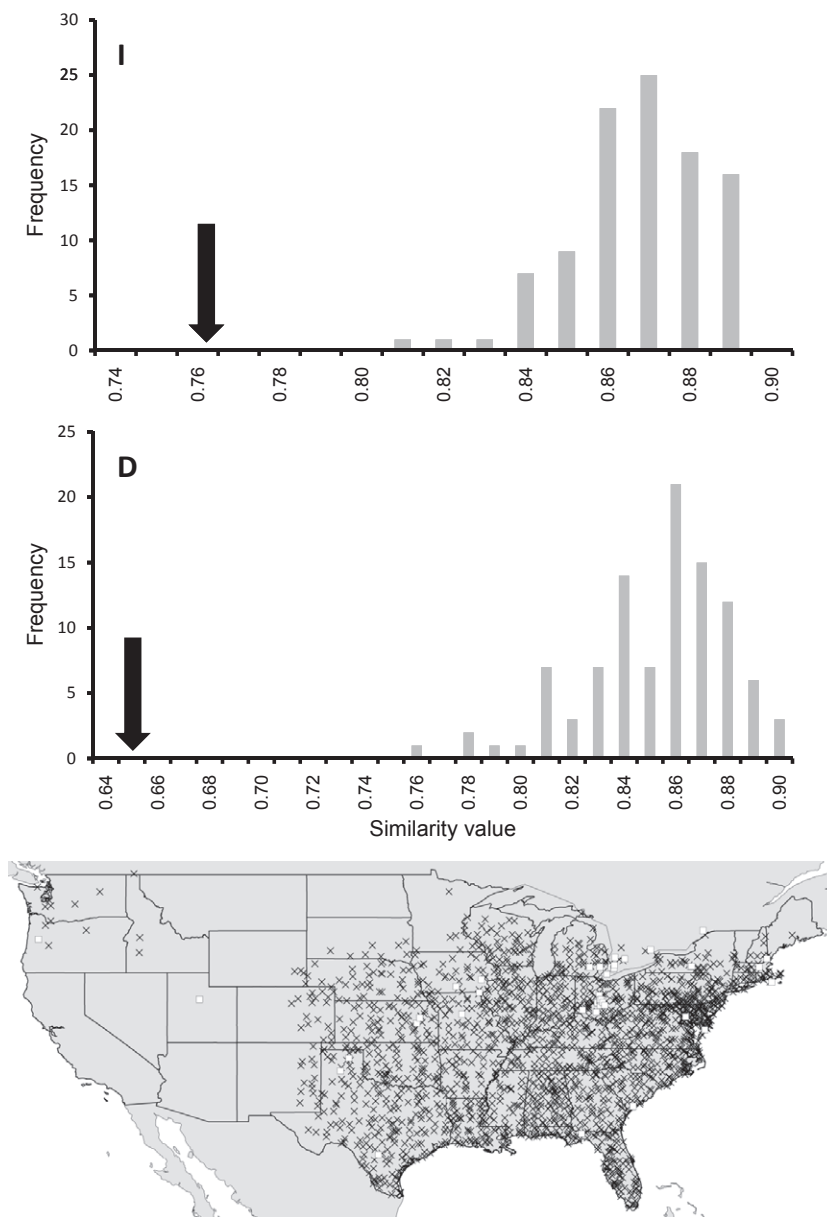
On submitting these occurrence points to the niche-identity and niche-similarity tests of Warren *et al.* (2008), differences between the tests of the two null hypotheses were at once evident. With regard to testing whether the niches of the two virtual species are more similar than would be expected by chance, as expected given that the occurrence data for the two species were sampled from the same subset of environmental space (i.e. my made-up ecological niche), the result was significant niche similarity (Fig. 2,  $P \leq 0.01$ ). However, in terms of niche *identity*, the niches of the two species were highly significantly differentiated (Fig. 2,  $P \leq 0.01$ ). Hence, although the two niches were indeed significantly similar, they were not identical.

As a second example of Type 1 error in niche-identity tests, I obtained two sets of data describing the distribution of the quail *Colinus virginianus*: 46 localities associated with natural history museum specimen collection sites (from various museum holdings via the ORNIS portal, <http://www.ornisnet.org>; accessed 15 December 2009), and 1845 North American Breeding Bird Survey (BBS, 2006) route detections (Fig. 3). I used Warren *et al.*'s (2008) niche-identity tests to assess



**Figure 2** Tests for niche identity and niche similarity (after Warren *et al.*, 2008) applied to two virtual species with distributions derived from a single hypothetical ecological niche (see map showing locations of the two species' ranges in North America in inset). *I* and *D* are two distinct measures of niche similarity (Warren *et al.*, 2008). Arrows indicate observed values, relative to the frequency distributions of random replicates from the niche-identity and niche-similarity tests.





**Figure 3** Illustration of Type 1 error in tests of niche identity, via two sets of data describing the distribution of the quail *Colinus virginianus* in North America. The top and middle panels show results of niche-identity tests: top, null distribution (histogram) and observed similarity value for the *I* statistic; middle, similar information for the *D* statistic (observed values shown by arrows). The bottom panel shows the distributions of the two sets of occurrence data: museum specimen locality data (circles) and North American Breeding Bird Survey data (crosses).

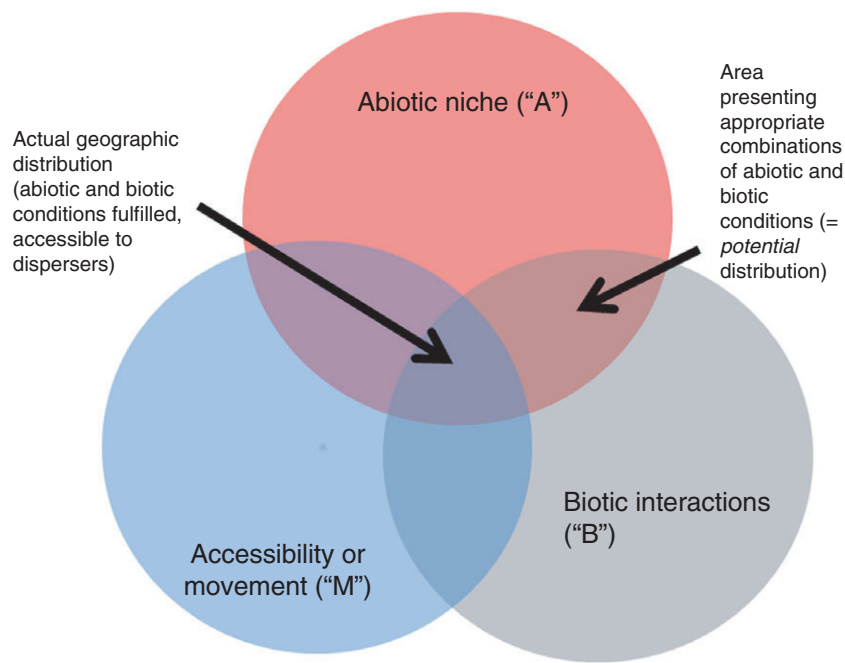
whether these two samples of points created models that were so dissimilar that it would be unlikely that they could have been drawn from the same population, and found that they also were significantly non-identical ( $P < 0.01$ ). Once again, a pair of samples that should produce identical niche models did not, probably owing to minutiae of sampling, and possibly owing to overfitting by the Maxent algorithm (see examples illustrated in Peterson *et al.*, 2007). For whatever reason, however, the niches of the two *C. virginianus* samples were not identical, again pointing towards a tendency to Type 1 errors in identity tests.

In sum, these results mirror the results that Warren *et al.* (2008) obtained in their re-analysis of the Peterson *et al.* (1999) data. That is, while they showed significant background similarity, the sister-species pairs frequently were significantly non-identical. I suspect that the niche-identity tests presented by Warren *et al.* (2008) are very sensitive, and will tend to

commit Type 1 errors, owing to variation in modelled niches resulting from different landscape characteristics, differences in model runs, and the vagaries of sampling. As a new generation of niche-based tools for exploring and testing these ideas emerges, then, it will be very important to focus on the biological significance of the conclusions, rather than on how well the results can be distinguished statistically. These results are mirrored almost exactly in a recent study by Godsoe (2010), which also concluded that niche-identity tests as indications of niche differentiation may frequently be confused by landscape-level differences in the environments represented.

### Need to control for availability and landscape constraints

One critical point that has become increasingly clear is the need to delimit areas for analysis with great care (Soberón &



**Figure 4** The 'BAM diagram', showing a simplified framework for understanding where species will and will not be distributed. Distributions of species are seen as responding to three sets of factors: the abiotic niche (A) and the biotic niche (B), which roughly correspond to the fundamental ecological niche (A) and the realized ecological niche ( $A \cap B$ , here termed the potential distribution). A further modification to distributional potential, however, is that of access (here M for 'movement'), which constrains species distributions quite dramatically.

Peterson, 2005), but no easy solutions have been offered. I refer to the 3-factor conceptual framework of Soberón & Peterson [2005; the 'BAM' (biotic, abiotic, movement) diagram] that considers biotic factors, abiotic factors and movement factors in delimiting geographical distributions of species (Fig. 4). Using this framework, the means by which one would ideally delimit the area of analysis becomes clear: the only areas 'sampled' by the species and either used or not used are those within its movement and dispersal possibilities, which is the 'M' in the BAM diagram. M is the area that is within the dispersal capabilities of the species in question, either in the present day or through the relevant past (perhaps since the Last Interglacial period, 135,000 years ago, in the case of many vertebrates).

As such, model development (and model evaluation as well) should be performed only across the area contained in M. Although estimating M is in itself not an easy task, as it is a concept that is variable as a function of time-scale, an explicit statement of assumptions regarding the extent of M is critical to generating high-quality models. In this way, contrasts between areas known to be inhabited and areas not known to be inhabited by the species in question are kept maximally relevant and meaningful.

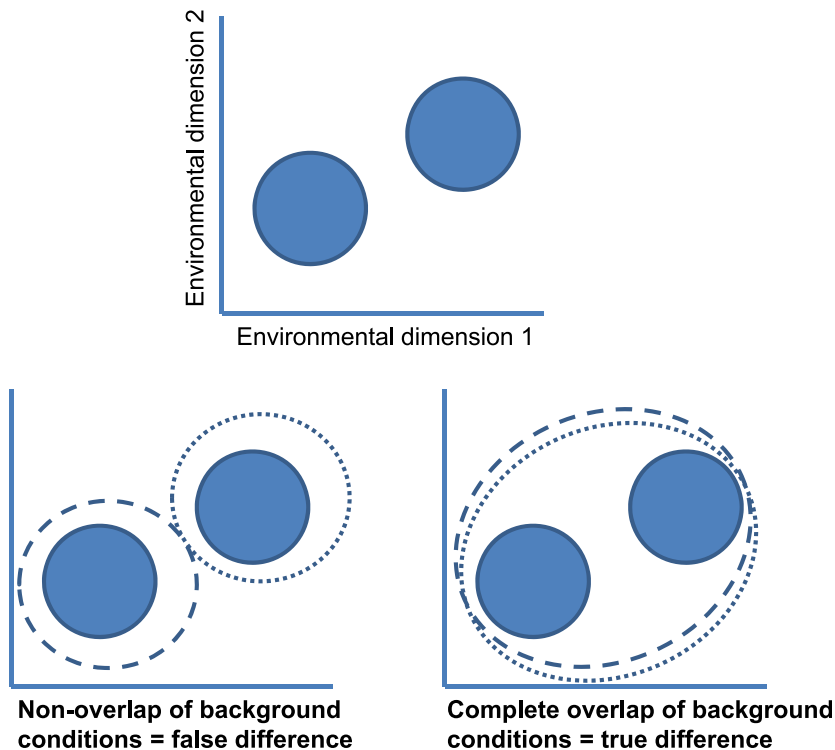
M is relevant not only to model development but also to model interpretation. Consider two niche models that are non-overlapping when visualized in environmental space (Fig. 5, top). Although the case for niche differentiation would appear to be clear, this conclusion is robust only when the conditions of the 'other' niche are within M for a given species (Fig. 5,

bottom right); that is, when the conditions used by the other species are available to, but not used by, the species of interest. In the opposite case (Fig. 5, bottom left), when the conditions used by the other species are not available to the species of interest, the conclusion of niche differentiation will not be robust (Kambhampati & Peterson, 2007). These considerations are similar to those necessary to understand the limits of predictions of the invasive potential of non-native species (Peterson, 2005). The Warren *et al.* (2008) background similarity tests are able to take into consideration the 'study area' for each species in determining whether niches are similar or not, which is ideally more or less equivalent to M.

## DISCUSSION

The field of 'species-level quantitative ecological biogeography' is young and will continue to evolve and mature over the coming decades. The debate over the question of niche conservatism is illustrative: some groups of researchers have concluded that conservatism is rampant, while other groups have concluded that niches evolve rather quickly and easily. While the difference of opinion is partly a result of methodological artefact (Warren *et al.*, 2008), it also reflects the immature nature of this field and the preliminary nature of the data and methods available, suggesting the need for synthesis and summary.

This review provides one step towards such a synthesis, taking a first time-structured look at niche conservatism and considering a much broader swathe of species than have



**Figure 5** Illustration of different interpretations of two non-overlapping niches (shaded shapes, top graph), depending on whether the surrounding conditions [i.e. in the area that is within the dispersal capabilities of the species (M), as shown by the dotted and dashed circles] are overlapping (bottom right) or non-overlapping (bottom left). In the case of overlapping background conditions, the conclusions of niche differentiation can be robust, but when background conditions are not overlapping, the conclusions of niche differentiation will not be well founded.

previous overviews. Araújo & Pearson (2005) and numerous other authors have pointed out a general state of equilibrium of distributions of species and climate dimensions. Although this conclusion has been debated in a recent high-profile publication (Beale *et al.*, 2008), the negative view appears to be based on the improper formulation of null models and interpretation of results (Araújo *et al.*, 2009; Peterson *et al.*, 2009). Still, a rich realm of further insight is that of more detailed analyses of niche characteristics and their change through time, as manifested across phylogeny – indeed, it is only over the phylogenetic history of relatively older groups that one observes frequent and marked niche change, so such studies offer the opportunity to understand the rate of niche change, the tempo and mode of niche change, and the correlates of such change events. This review thus clarifies the climate-related discussion by assessing at a very coarse temporal resolution the longevity and constancy of climate–distribution associations across a broad sample of species.

Pearman *et al.* (2008) noted that ‘niche conservatism’ is achieved only by simultaneous non-change in all of the three BAM factors (biotic, abiotic, movement), whereas non-conservatism is achieved by any change in any of the three factors. Indeed, non-conservatism can take the form of change in the niche cloud via (1) a shift in position in environmental space, (2) a change in shape, or (3) a change in volume (Jackson & Overpeck, 2000). As a result of this review, niche conservatism can be seen to be highly dependent on the time-scale of the comparison, the resolution and extent of the analysis, and the details of model fitting and comparisons, particularly as regards the null hypotheses being tested (Warren *et al.*, 2008).

### Niche conservatism and speciation

Peterson *et al.* (1999) presented evidence that speciation is not typically accompanied by ecological innovation, but Graham *et al.* (2004) argued the opposite. This contrast was clarified greatly by Warren *et al.* (2008), who pointed out that the two studies had tested distinct null hypotheses, and probably for that reason had arrived at opposite conclusions. That is, Peterson *et al.* (1999) used tests of significance of niche similarity, whereas Graham *et al.* (2004) used tests of significance of niche identity, with consequently differing results. Warren *et al.* (2008), however, given their re-analysis of the original data, went on to state that Peterson *et al.*’s (1999) assertion of niche conservatism during speciation ‘is further challenged by the rejection of the hypothesis of niche equivalency’; hence, Warren *et al.* (2008) went beyond the demonstration of niche non-identity to make biological conclusions regarding the ecological context of speciation events. However, the niche-identity test of Warren *et al.* (2008) is put into perspective by the heuristic analyses offered in this review, in which occurrence points from two distributional areas chosen from a single artificial niche or two samples of occurrences from the same species were shown to be significantly non-identical. Warren *et al.* (2008) did offer the caveat that ‘the probabilities obtained would accurately reflect the probabilities under the null hypothesis if the sampling of each species was unbiased with respect to its environmental tolerances’. I suggest that such biases may be rampant among real analyses of real data on real species, and that biases may even have entered the picture when random points were generated in reasonably small numbers from my virtual species. The high dimension-



ality of typical environmental spaces in niche-modelling exercises may interact with biases introduced by the M (movement) factors in the BAM diagram, which in the Warren *et al.* (2008) niche-identity tests cannot be controlled. Furthermore, the tendency on the part of some algorithms used to estimate niches to overfit to sampling biases (Peterson *et al.*, 2007) may contribute to this phenomenon. In short, I suspect that niche-identity tests are detecting very real – but biologically non-significant – differences in the ecological ‘niche’ of the combination of species  $\times$  landscape  $\times$  sampling; as such, these tests will turn out to be very prone to Type 1 errors and will over-represent niche differentiation if not interpreted carefully.

In light of the above discussions, and particularly in view of the temporally structured review of niche conservatism that I have developed, it appears that speciation is rarely accompanied by dramatic Grinnellian niche evolution. [Note that finer-scale ‘bionomic’ dimensions of niche are little studied in speciation, and may indeed change (Soberón, 2007).] The fascinating arrays of factors that may constrain niche plasticity are explored by Pelini *et al.* (2009). Rather, niche conservatism appears to break down more on the scale of diversification of species within genera; that is, as lineages diversify from pairs of species into more diverse clades. This observation shifts the evolution of ecological traits farther into the past, and away from the time frame of speciation events. As such, as in the case of niche differentiation during species invasions and its alleged demonstrations discussed above (Peterson & Nakazawa, 2008), I suggest that such cases will be rare, and that demonstrations of ecological innovation during speciation events must be documented extremely carefully, so as to avoid mistaken conclusions.

## Ways forward

This review points out the critical, if obvious, role of time in the consideration of niche conservatism. While the point may *seem* obvious, it has been neglected. Only relatively recently, with the advent of phylogenetic studies of niche conservatism or non-conservatism (e.g. Knouft *et al.*, 2006; Eaton *et al.*, 2008), has the temporal scale of this phenomenon been considered in detail (see the intriguing analysis of order of trait appearance in Ackerly *et al.*, 2006). As a result, an important general recommendation for future progress is that of the explicit consideration of temporal dimensions in studies of ecological niche evolution.

To be able to speak effectively about niche conservatism in the first place, however, it is important to ensure that inferences of niche characteristics and their similarity among lineages are robust, and not vulnerable to biases. For this reason, careful control of model dimensionality and complexity is crucial in model training, as careless handling of these issues can result in niche estimates that *appear* to be distinct among species or samples when such is not necessarily the case. These points are important for ensuring that any conclusions of niche difference are genuinely

attributable to niche difference, and not to methodological artefact.

A further consideration, much more biological and biogeographical in nature, is that of identifying regions relevant to analyses. Species do not select their ranges from across the full universe of possible environments and regions; rather, they are constrained to occur in areas that are within their dispersal potential from their areas of origin (Pulliam, 2000; Soberón & Peterson, 2005). As a consequence, estimating this area of dispersal ‘reach’ and constraining analyses and comparisons to within that area and/or to within that set of environments are both crucial. This context – in essence, the set of environments that corresponds to the area within the M (accessibility or mobility) constraint of the BAM framework (Soberón & Peterson, 2005) – identifies the set of areas and environmental conditions that are at all *relevant* to the question of niche similarity or difference. As recently emphasized in the design of tools offered by Warren *et al.* (2008), and as has been clear in recent re-analyses of previous research results (see, for example, the case of Mexican jays and the different conclusions reached by Peterson & Holt, 2003; Rice *et al.*, 2003; and McCormack *et al.*, 2010), this explicit consideration of dispersal abilities of species over relevant periods of time can change interpretations of niche similarity or difference radically.

Finally, the results of this analysis suggest that the implications of niche conservatism should be fully integrated into evolutionary biology and biogeography (Knowles *et al.*, 2007; Peterson, 2009; Waltari & Guralnick, 2009): pervasive niche conservatism enables many features of the potential geography of species to be reconstructed with greater confidence, providing exciting new insights (Peterson, 2009). These results also serve to focus interest and attention on the apparently rare situations in which coarse-resolution, scenopoetic ecological niches *do change*, as these situations may be highly unusual and thus insightful.

## ACKNOWLEDGEMENTS

I thank many colleagues across this emerging field, particularly Jorge Soberón, Enrique Martínez-Meyer, Rob Guralnick, Rob Anderson, Richard Pearson, Miguel Araújo, Miguel Nakamura, J. Alexandre Diniz-Filho, Alberto Jiménez-Valverde, Dan Warren, and many others with whom I have discussed these issues. Funding was provided by Microsoft Research. This paper stems from the conference *Niche Evolution* held in Zurich on 3–4 July 2009.

## REFERENCES

- Ackerly, D.D. (2003) Community assembly, niche conservatism, and adaptive evolution in changing environments. *International Journal of Plant Sciences*, **164**, S165–S184.
- Ackerly, D.D., Schilck, D.W. & Webb, C.O. (2006) Niche evolution and adaptive radiation: testing the order of trait divergence. *Ecology*, **87**, S50–S61.

- Araújo, M.B. & Pearson, R.G. (2005) Equilibrium of species' distributions with climate. *Ecography*, **28**, 693–695.
- Araújo, M.B., Pearson, R.G., Thuiller, W. & Erhard, M. (2005) Validation of species–climate impact models under climate change. *Global Change Biology*, **11**, 1504–1513.
- Araújo, M.B., Thuiller, W. & Yoccoz, N.G. (2009) Reopening the climate envelope reveals macroscale associations with climate in European birds. *Proceedings of the National Academy of Sciences USA*, **105**, e45–e46.
- BBS (2006) *North American Breeding Bird Survey*. US Geological Survey, Washington, DC. Available at: <http://www.mp2-pwrc.usgs.gov/bbs> (accessed 15 December 2009).
- Beale, C.M., Lennon, J.J. & Gimona, A. (2008) Opening the climate envelope reveals no macroscale associations with climate in European birds. *Proceedings of the National Academy of Sciences USA*, **105**, 14908–14912.
- Benedict, M.Q., Levine, R.S., Hawley, W.A. & Lounibos, L.P. (2007) Spread of the tiger: global risk of invasion by the mosquito *Aedes albopictus*. *Vector-Borne and Zoonotic Diseases*, **7**, 76–85.
- Broennimann, O. & Guisan, A. (2008) Predicting current and future biological invasions: both native and invaded ranges matter. *Biology Letters*, **4**, 585–589.
- Broennimann, O., Treier, U.A., Müller-Schärer, H., Thuiller, W., Peterson, A.T. & Guisan, A. (2007) Evidence of climatic niche shift during biological invasion. *Ecology Letters*, **10**, 701–709.
- Eaton, M.D., Soberón, J. & Peterson, A.T. (2008) Phylogenetic perspective on ecological niche evolution in American blackbirds (family Icteridae). *Biological Journal of the Linnean Society*, **94**, 869–878.
- Elith, J., Graham, C.H., Anderson, R.P. *et al.* (2006) Novel methods improve prediction of species' distributions from occurrence data. *Ecography*, **29**, 129–151.
- Fitzpatrick, M.C., Weltzin, J.F., Sanders, N.J. & Dunn, R.R. (2007) The biogeography of prediction error: why does the introduced range of the fire ant over-predict its native range? *Global Ecology and Biogeography*, **16**, 24–33.
- Godsoe, W. (2010) Regional variation exaggerates ecological divergence in niche models. *Systematic Biology*, **59**, 298–306.
- Graham, C.H., Ron, S.R., Santos, J.C., Schneider, C.J. & Moritz, C. (2004) Integrating phylogenetics and environmental niche models to explore speciation mechanisms in dendrobatid frogs. *Evolution*, **58**, 1781–1793.
- Hijmans, R.J., Cameron, S.E., Parra, J.L., Jones, P.G. & Jarvis, A. (2005) Very high resolution interpolated climate surfaces for global land areas. *International Journal of Climatology*, **25**, 1965–1978.
- Hirzel, A.H., Helfer, V. & Metral, F. (2001) Assessing habitat-suitability models with a virtual species. *Ecological Modelling*, **145**, 111–121.
- Huntley, B., Bartlein, P.J. & Prentice, I.C. (1989) Climatic control of the distribution and abundance of beech (*Fagus L.*) in Europe and North America. *Journal of Biogeography*, **16**, 551–560.
- Jackson, S.T. & Overpeck, J.T. (2000) Responses of plant populations and communities to environmental changes of the late Quaternary. *Paleobiology*, **26**(Supplement), 194–220.
- Kambhampati, S. & Peterson, A.T. (2007) Ecological niche conservation and differentiation in the wood-feeding cockroaches, *Cryptocercus*, in the United States. *Biological Journal of the Linnean Society*, **90**, 457–466.
- Knouft, J.H., Losos, J.B., Glor, R.E. & Kolbe, J.J. (2006) Phylogenetic analysis of the evolution of the niche in lizards of the *Anolis sagrei* group. *Ecology*, **87**, S29–S38.
- Knowles, L.L., Carstens, B.C. & Keat, M.L. (2007) Coupling genetic and ecological-niche models to examine how past population distributions contribute to divergence. *Current Biology*, **17**, 940–946.
- Lobo, J.M., Jiménez-Valverde, A. & Real, R. (2008) AUC: a misleading measure of the performance of predictive distribution models. *Global Ecology and Biogeography*, **17**, 145–151.
- Losos, J.B. (2008) Phylogenetic niche conservatism, phylogenetic signal and the relationship between phylogenetic relatedness and ecological similarity among species. *Ecology Letters*, **11**, 995–1003.
- Losos, J.B., Leal, M., Glor, R.E., de Queiroz, K., Hertz, P.E., Rodríguez Schettino, L., Chamizo Lara, A., Jackman, T.R. & Larson, A. (2003) Niche lability in the evolution of a Caribbean lizard community. *Nature*, **424**, 542–545.
- Martínez-Meyer, E. & Peterson, A.T. (2006) Conservatism of ecological niche characteristics in North American plant species over the Pleistocene-to-Recent transition. *Journal of Biogeography*, **33**, 1779–1789.
- McCormack, J.E., Zellmer, A.J. & Knowles, L.L. (2010) Does niche divergence accompany allopatric divergence in *Aphelocoma* jays as predicted under ecological speciation? Insights from tests with niche models. *Evolution*, **65**, 184–202.
- Medley, K.A. (2010) Niche shifts during the global invasion of the Asian tiger mosquito, *Aedes albopictus* Skuse (Culicidae), revealed by reciprocal distribution models. *Global Ecology and Biogeography*, **19**, 122–133.
- Pearman, P.B., Guisan, A., Broennimann, O. & Randin, C.F. (2008) Niche dynamics in space and time. *Trends in Ecology and Evolution*, **23**, 149–158.
- Pelini, S.L., Dzurisin, J.D.K., Prior, K.M., Williams, C.M., Marisco, T.D., Sinclair, B.J. & Hellmann, J.J. (2009) Translocation experiments with butterflies reveal limits to enhancement of poleward populations under climate change. *Proceedings of the National Academy of Sciences USA*, **106**, 11160–11165.
- Peterson, A.T. (2003) Predicting the geography of species' invasions via ecological niche modeling. *Quarterly Review of Biology*, **78**, 419–433.
- Peterson, A.T. (2005) Predicting potential geographical distributions of invading species. *Current Science*, **89**, 9.
- Peterson, A.T. (2007) Why not WhyWhere: the need for more complex models of simpler environmental spaces. *Ecological Modelling*, **203**, 527–530.

- Peterson, A.T. (2009) Phylogeography is not enough: the need for multiple lines of evidence. *Frontiers in Biogeography*, **1**, 19–25.
- Peterson, A.T. & Holt, R.D. (2003) Niche differentiation in Mexican birds: using point occurrences to detect ecological innovation. *Ecology Letters*, **6**, 774–782.
- Peterson, A.T. & Nakazawa, Y. (2008) Environmental data sets matter in ecological niche modelling: an example with *Solenopsis invicta* and *Solenopsis richteri*. *Global Ecology and Biogeography*, **17**, 135–144.
- Peterson, A.T. & Nyári, Á. (2007) Ecological niche conservatism and Pleistocene refugia in the thrush-like mourner, *Schiffornis* spp., in the Neotropics. *Evolution*, **62-I**, 173–183.
- Peterson, A.T. & Shaw, J.J. (2003) *Lutzomyia* vectors for cutaneous leishmaniasis in southern Brazil: ecological niche models, predicted geographical distributions, and climate change effects. *International Journal of Parasitology*, **33**, 919–931.
- Peterson, A.T. & Vargas-Barajas, N. (1993) Ecological diversity in scrub jays, *Aphelocoma coerulescens*. *Biological diversity of Mexico: origins and distribution* (ed. by T.P. Ramamoorthy, R. Bye, A. Lot and J. Fa), pp. 309–317. Oxford University Press, New York.
- Peterson, A.T., Soberón, J. & Sánchez-Cordero, V. (1999) Conservatism of ecological niches in evolutionary time. *Science*, **285**, 1265–1267.
- Peterson, A.T., Papeş, M. & Eaton, M. (2007) Transferability and model evaluation in ecological niche modeling: a comparison of GARP and Maxent. *Ecography*, **30**, 550–560.
- Peterson, A.T., Papeş, M. & Soberón, J. (2008) Rethinking receiver operating characteristic analysis applications in ecological niche modelling. *Ecological Modelling*, **213**, 63–72.
- Peterson, A.T., Barve, N., Bini, L.M., Diniz-Filho, J.A., Jiménez-Valverde, A., Lira-Noriega, A., Lobo, J.M., Maher, S., Marco Júnior, P.D., Martínez-Meyer, E., Nakazawa, Y. & Soberón, J. (2009) The climate envelope may not be empty. *Proceedings of the National Academy of Sciences USA*, **106**, E47.
- Pulliam, H.R. (2000) On the relationship between niche and distribution. *Ecology Letters*, **3**, 349–361.
- Raxworthy, C.J., Ingram, C.M., Rabibisoa, N. & Pearson, R.G. (2007) Applications of ecological niche modeling for species delimitation: a review and empirical evaluation using day geckos (*Phelsuma*) from Madagascar. *Systematic Biology*, **56**, 1–17.
- Rice, N.H., Martínez-Meyer, E. & Peterson, A.T. (2003) Ecological niche differentiation in the *Aphelocoma* jays: a phylogenetic perspective. *Biological Journal of the Linnean Society*, **80**, 369–383.
- Ricklefs, R.E. & Latham, R.E. (1992) Intercontinental correlation of geographical ranges suggests stasis in ecological traits of relict genera of temperate perennial herbs. *The American Naturalist*, **139**, 1305–1321.
- Rödger, D. & Lötters, S. (2009) Niche shift versus niche conservatism? Climatic characteristics of the native and invasive ranges of the Mediterranean house gecko (*Hemidactylus turcicus*). *Global Ecology and Biogeography*, **18**, 674–687.
- Soberón, J. (2007) Grinnellian and Eltonian niches and geographic distributions of species. *Ecology Letters*, **10**, 1115–1123.
- Soberón, J. & Peterson, A.T. (2005) Interpretation of models of fundamental ecological niches and species' distributional areas. *Biodiversity Informatics*, **2**, 1–10.
- Vanreusel, W., Maes, D. & Van Dyck, H. (2007) Transferability of species distribution models: a functional habitat approach for two regionally threatened butterflies. *Conservation Biology*, **21**, 201–212.
- Waltari, E. & Guralnick, R.P. (2009) Ecological niche modeling of montane mammals in the Great Basin, North America: examining past and present connectivity of species across basins and ranges. *Journal of Biogeography*, **36**, 148–161.
- Warren, D.L., Glor, R.E. & Turelli, M. (2008) Environmental niche equivalency versus conservatism: quantitative approaches to niche evolution. *Evolution*, **62**, 2868–2883.
- Wiens, J.J. (2008) Commentary on Losos (2008): niche conservatism déjà vu. *Ecology Letters*, **11**, 1004–1005.
- Wiens, J.J. & Graham, C.H. (2005) Niche conservatism: integrating evolution, ecology, and conservation biology. *Annual Review of Ecology, Evolution, and Systematics*, **36**, 519–539.

## SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this article:

**Appendix S1** Summary of studies from the scientific literature that were reviewed in the course of the analyses presented in this paper.

As a service to our authors and readers, this journal provides supporting information supplied by the authors. Such materials are peer-reviewed and may be re-organized for online delivery, but are not copy-edited or typeset. Technical support issues arising from supporting information (other than missing files) should be addressed to the authors.

## BIOSKETCH

**A. Townsend Peterson** is University Distinguished Professor in Ecology and Evolutionary Biology, and Curator in the Biodiversity Institute, at the University of Kansas. His research focuses on the geography, history and ecology of species distributions, and ranges from the spatial dimensions of disease transmission to the historical biogeography of tropical birds.

Editor: Peter B. Pearman